

The assignment set consists of 3 pages

KRISTIANIA

PG3302 Software design

Exam/semester assignment

Implementation

- **The assignment should be submitted in groups of 1-2 persons**, via **WISEflow**.
 - **You are allowed to look at all the written material related the course** (slides, textbook, own answers, and suggested exercise solutions on Canvas.)
 - It is **not allowed** to seek **help from people outside the group**. (*Exception: You can ask your teacher for clarifications.*)
 - It is **not allowed** to ask **artificial intelligence to write the code** for you (outside of an IDEs built-in CoPilot or the like).
 - The hand-in **must contain**:
 1. **PDF document** describing the process and content/use of the solution.
 2. **Zip with source code** and any other necessary files in that context.
 3. **If possible: (not required)** Zip with executable version (.exe file and related folders) for Windows 11.
 - **Scope:** It is expected to use **approx. 30 hours of work per person**. This is calculated as follows: 1 person will contribute on average 8 hours per full week to this assignment, for an exam duration of 3.7 consecutive weeks. (But, of course, you can portion out the work in another way. For example: Do it all at continuously by working 30 hours in a single week.)
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Delivery info for the 3 parts

Here follows some more information about the 3 parts in the delivery:

1. PDF document describing the process and content/use of the solution

This part involves writing a pdf report documenting the content of the project you have worked on.

Note: This is an important part! Among other things, this is how you tell the assessor what parts of the syllabus you have covered. Make sure to include screenshots of the running program, your code, and your design sketches where it enhances the understanding or quality of the documentation.

Elements to include in the pdf document:

1. **The process:** Write a paragraph or two about how you worked on your hand-in. If you wish to remain anonymous, do not write your name, instead use: "person 1", "person 2", etc.
2. **Which parts of the syllabus you have included**, and how they are used. Make sure you document the extent to which you have used **the following 10 syllabus topics**:
 1. Techniques and tools for version control and collaboration (the latter only relevant in groups of 2, with techniques such as pair programming).
 2. UML class diagram for your solution.
 3. C# .NET's modern features: properties, data types with '?' at the end, use of the object class's methods/properties, perhaps operator overloading?

4. Description of the SOLID principles: which you followed and how this can be seen in your code.
 5. Layering.
 6. Extensive description of how you did data storage, what framework you chose for this, and why you did it that way. Available data storage options: SQLite database, JSON file format, XML file format. (Make sure your hand-in has pre-generated data in it.)
 7. Design patterns (which ones, and where in the code they are used).
 8. Testing: Unit tests, integration tests, maybe mocks? TDD or not. *Note:* Unit tests that cover an entire solution are very comprehensive to write. Therefore, it is ok if you only use it on parts of your solution. If so, describe what you choose to test. Implement tests so that you can show the breadth of what you have learned regarding testing.
 9. Refactoring. (Not always easy to spot in the code afterwards, must be well described.)
 10. Anything else you wish to specify that could up the assessment/evaluation?
3. **Known bugs:** Do you know of any errors or deficiencies in the program? Name them here.
 4. **Source reference, IMPORTANT:** If it is discovered that you have used ready-made material that you do not disclose (do not mention here), it is considered cheating! Therefore, state your sources, we want to avoid cases of plagiarism!

Remember 1: We have learned how to model programming projects (code) in this subject. Figures and diagrams you produce in that context should be handed in. It can be screenshots from digital tools, photos of paper-based drawings, etc.

Remember 2: Software design is as much about the development of a program as the finished program. Make sure you document both the solution *and* the process well!

2. Zip with source code and other necessary files

This is the code for the program. Provide the source code (and associated files, like files for data storage), so that the assessor can see the programming and code structure! Must be openable in Visual Studio 2022.

Important: The assessor cannot be expected to install tools other than those demonstrated in the lessons. (NuGet packages are fine.) So, if you use a DB for storage (which is a good/advanced choice), use *SQLite* as it doesn't require any installation of software on the assessor's side.

3. *If possible: (not required)* Zip with executable version (.exe file and related folders) for Windows 11

The executable version should, if possible, be available for Windows 11. Make sure that both the .exe file and any necessary folders are included. If you do not have the possibility of building a version that runs on Windows, explain why ("team did development on macOS") and make sure the content in point 2 (above) is in a state that lets the assessor build an executable Win 11 version.

Requirements regarding content

This exam involves **designing a software solution**, then **programming that software solution in C#**.

Design a software solution

When it comes to "designing a software solution" the code structure is the primary focus. But the user experience is also important (not the visuals/graphics of the console front-end, but the useability), so the solution should be intuitive and self-explanatory/well documented.

Programming the software solution in C#

The solution should at least cover the **standard 3-layer architecture**. Therefore:

1. Data storage at the lower layer (e.g., storage to SQLite DB, JSON file or XML file).
2. Domain logic in the middle. You are free to choose what domain the program will handle.
3. Console-based front end. (Not much focus on graphical appearance, it is not part of the syllabus. But there should be a logical and user-friendly front end, it is part of planning and design.)

It is expected that the hand-in contains at least 2 projects:

- C# Console App
- C# NUnit Test Project

Note: Think good structure, not necessarily what is practical for such a small assignment outside of a school setting. Include most elements from the syllabus (see sections 2.1 - 2.10 under pdf document, above). If you do something that doesn't seem logical, explain why you include it, both as a code comment and in your attached pdf document.

Examples and inspiration for documentation and domain

Regarding the pdf report, you can find 2 example reports on Canvas (one in English and one in Norwegian), in the PG3302 exam module.

Regarding the domain for your hand-in, any traditional business-to-consumer setup would be a good choice. It should fulfil all the following:

- A system that uses the full range of CRUD commands.
- Store data to an SQLite DB (advanced) or to a JSON/XML file (more basic) as part of the backend.
- Have a menu in the console frontend where the user chooses operations and see results. While it doesn't have to be graphically pretty, it should be user friendly and well-designed regarding its tasks.

Note: Making a game is *not* recommended. While it is fun to work on such a setup, it makes it harder to show the full width of the syllabus.

- End of task set -